# Diaz Rifqi Munggaran

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## Education

**Siliwangi University** Tasikmalaya, West Java
*Bachelor's of Science, Informatics* *Aug 2022 — Jul 2026*

- Cumulative GPA: 3.71/4.0
- Relevant Coursework: Relational Databases, Data Structures, Program Development, Discrete Mathematics, Linear Algebra, Statistics, Machine Learning, Deep Learning, Computer Vision

## Work Experience

**Data Analyst Intern** Apr 2025 — Jun 2025
Badan Pusat Statistik, Kota Tasikmalaya *Tasikmalaya, West Java*

- Performed data entry, cleaning, and standardization for multiple official price surveys (SHP, SHK, SHBP) in BPS's web-based system, ensuring consistent data for downstream statistical analysis and reporting.
- Combined and prepared data from multiple sources (SILINDA and BAPANAS) to build an LSTM time series model that forecasts rice prices, achieving MAPE as low as 0.06% on key categories.
- Contributed to backend development for a standalone website: managing data, integrating the prediction model into the system, and storing 1,000-day forward forecasts.
- Supported UMKM (SME) profiling with the MatchaPro application: data processing, verification, and mapping by business sector and location.
- Presented project progress and model evaluation results regularly to BPS supervisors.

## Projects

**Intern**, Rice Price Prediction (LSTM) — BPS Kota Tasikmalaya Apr 2025 — Jun 2025

- Designed, trained, and evaluated an LSTM forecasting model on SILINDA and BAPANAS data, then integrated it into a backend system to serve long-horizon predictions.

**Workshop (Bengkel) Customer Management — ERD Design**

- Designed an Entity Relationship Diagram (ERD) for a workshop customer management system, modeling the entities and relationships needed to store and manage customer data

**Researcher (Undergraduate Thesis)**, YOLOv8 + ResGAM Football Player Detection Jan 2026 — Jul 2026

- Modified YOLOv8 by integrating a custom ResGAM module (Residual Block + Global Attention Mechanism) to improve football player detection
  - Improved mAP@0.5 from 0.722 to 0.825 and F1-score from 0.73 to 0.851 on the DFL Bundesliga Data Shootout dataset
- Built dataset splitting, augmentation, and class-balancing pipelines to address severe class imbalance (774 ball instances vs. 17,865 player instances)
- Benchmarked four YOLO variants (YOLOv5, YOLOv8, YOLO11, YOLO26) and documented the results in a thesis and a parallel journal manuscript

**Developer**, Football Analysis Pipeline Jan 2026 — Jul 2026

- Built an end-to-end pipeline combining detection, ByteTrack tracking, team classification (SigLIP + UMAP + KMeans), and homography-based pitch projection
- Migrated the pipeline from a hosted inference API to local model loading and resolved dependency conflicts

## Skills

- **Programming**: Python
- **Databases**: SQL, MySQL, PostgreSQL (relational); MongoDB (non-relational); ERD Design
- **Data**: Data Cleaning, Data Preprocessing, Data Integration, Time Series Forecasting (LSTM)
- **Machine Learning**: Deep Learning, Computer Vision, PyTorch, Ultralytics (YOLO), OpenCV, Hugging Face, Roboflow
- **Tools**: Git, Git LFS, Google Colab, Microsoft Excel
- **Networking**: MikroTik (MTCNA certified)

- **Documentation**: Technical writing (thesis, journal manuscript, project reports and presentations)

## Certificates

**MikroTik Certified Network Associate (MTCNA)**, MikroTik May 2024 — May 2027
- Credential ID: 2405NA0537

**Test of English Language Proficiency (TELP)**, Siliwangi University Language Centre May 2024 — May 2026
- Score: 625